

Spectral Metrics of Transformer Training Are Determined by a Single Direction

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Paper under double-blind review

Abstract

Effective rank, RankMe, the leading eigenvalue share E_1 , CKA and anisotropy are used to describe how transformer representations change during pretraining. We show that one direction fixes the value of the rank statistics among them. For any statistic that takes the exponential entropy of a normalized spectrum, removing the top eigendirection $\mathbf{s}$ from the hidden states $\mathbf{H}$ gives the identity $M(\mathbf{H}) = e^{h_b(t)} M(\mathbf{C})^{1-t}$, where $\mathbf{C}$ is the residual and t is the share of the spectrum that $\mathbf{s}$ carries. It reproduces effective rank to 3×10^{-15} relative error on all 4,743 measurements we hold: Pythia at 160M, 410M and 1.4B over 20 checkpoints, OLMo-2 1B over 13, every layer of each, three corpus blocks, plus a sample-size sweep. Because effective rank normalizes eigenvalues while RankMe normalizes singular values, the two receive different shares, and we prove that RankMe’s is never the larger of the two. That ordering decides the damage: at the peak layer of Pythia-1.4B a single direction holds 85 percent of the eigenvalue mass but 6 percent of the singular-value mass, and distorts the two statistics by $243\times$ and $1.24\times$. The direction is the attention sink, which acquires that role during training, ending with roughly half its energy on the $1/512$ of tokens at position 0; its strength alone accounts for 94 to 97 percent of the variance in E_1 . Projecting it out turns a $105\times$ collapse in effective rank into $2.1\times$ growth, leaves the raw RankMe trajectory pointing the wrong way at two of three scales, and leaves one early transient that OLMo-2 reproduces. OLMo-2 grows no sink over its first 76B tokens, and there the raw and residual metrics differ by a median of 3.5 percent against $32\times$ to $243\times$ in Pythia, which separates the sink from the projection used to study it. Because entropy-based ranks have not converged at 524,288 tokens, we report ratios at a fixed token count and never an absolute rank.

1 Introduction

Effective rank, RankMe, the leading eigenvalue share E_1 , CKA and anisotropy are computed from one $n \times d$ matrix of token representations, and all of them have been read as descriptions of how a transformer organizes information over training. Li et al. (2025) use RankMe and a power-law spectral exponent to report three geometric phases in pretraining. The individual statistics were established earlier: anisotropy by Ethayarajh (2019), rogue dimensions by Timkey & van Schijndel (2021), RankMe as a label-free proxy for downstream quality by Garrido et al. (2023), and layer comparison by CKA in Raghu et al. (2021).

Those same matrices are the subject of a second literature, which reports that a few coordinates carry variance orders of magnitude above the rest. Xiao et al. (2024) named the attention sink and Sun et al. (2024) measured the massive activations behind it. Gu et al. (2025) traced when the sink appears during training, and Queipo-de Llano et al. (2025) proved that massive activations force the effective rank of a layer down. The two literatures rarely cite each other.

Problem. Fix a checkpoint, a layer, and a corpus of n tokens, and let $\mathbf{H} \in \mathbb{R}^{n \times d}$ hold the hidden states. Let $M(\mathbf{H})$ be a spectral summary. Given a trajectory $\{M(\mathbf{H}_t)\}$ over training steps t , decide which features of the trajectory describe the distribution of the n representations and which describe the size of a few outlier coordinates.

Write $\mathbf{s}$ for the top eigenvector of the covariance of $\mathbf{H}$, and $\mathbf{C} = \mathbf{H}(\mathbf{I} - \mathbf{s}\mathbf{s}^\top)$ for the residual after $\mathbf{s}$ is projected out. Every statistic in the list above that takes the exponential entropy of a normalized spectrum satisfies

$$M(\mathbf{H}) = e^{h_b(t)} \cdot M(\mathbf{C})^{1-t}, \quad (1)$$

with h_b the binary entropy in nats and t the share of that spectrum carried by $\mathbf{s}$. This is an identity. It holds for every matrix and constrains nothing about $\mathbf{C}$. A study that reports effective rank falling from 442 to 4.2 has reported t and $M(\mathbf{C})$, and nothing else.

Which t enters depends on the statistic. Effective rank normalizes eigenvalues, so $t = E_1 = \lambda_1 / \sum_i \lambda_i$; RankMe normalizes singular values, so $t = \sigma_1 / \sum_i \sigma_i$. Lemma 2 shows the second is never the larger of the two. One direction can therefore ruin effective rank and leave RankMe close to its residual value, which is the split Section 5 finds in Pythia. The residue left in RankMe is small. It is also enough to reverse the sign of the RankMe trajectory at two of the three scales we measured.

Contributions.

1. Section 3 proves the identity for effective rank and RankMe, decomposes anisotropy, bounds CKA’s homogenization, and proves the ordering $r \leq p$ between the two rank shares. The effective-rank instance reproduces measurement to 3×10^{-15} on 4,743 records and the ordering is never violated.
2. The ordering predicts a split that Table 1 confirms. One direction at the peak layer of Pythia-1.4B takes 85 percent of the eigenvalue mass and 6 percent of the singular-value mass, and distorts the two statistics by $243\times$ and $1.24\times$.
3. The distortion ratio is bounded above by the residual value, which grows with model width. Distortion factors quoted for models of different sizes are therefore not comparable, and the $700\times$ figures in the literature are partly a statement about d .
4. Across Pythia at three scales the direction ends up carrying about half its energy on the 1/512 of tokens at position 0, and its strength alone accounts for 94 to 97 percent of the variance in E_1 . Raw effective rank falls by $21\times$ to $119\times$ while the residual grows.
5. OLMo-2 1B forms no sink in its first 76B tokens. There raw and residual metrics differ by a median of 3.5 percent, against $32\times$ to $243\times$ in Pythia, which is the control the argument needs: the gap follows sink strength and not the projection.

The propositions in Section 3 are elementary. Their consequence is not. A class of published measurements turns out to carry no information beyond two other numbers, and one of the five statistics survives the problem far better than the others.

2 Setting

Hidden states. Fix a model, a layer, and a corpus. Running the corpus through the model yields n token representations in $\mathbb{R}^d$, stacked as the rows of $\mathbf{H} \in \mathbb{R}^{n \times d}$. Write $\bar{\mathbf{H}} = (\mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top)\mathbf{H}$ for the column-centered matrix, $\Sigma = \frac{1}{n}\bar{\mathbf{H}}^\top\bar{\mathbf{H}}$ for the covariance, and $\lambda_1 \geq \dots \geq \lambda_d \geq 0$ for its eigenvalues with eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_d$. Set $V = \text{tr } \Sigma$ and $\sigma_i = \sqrt{n\lambda_i}$ for the singular values of $\bar{\mathbf{H}}$.

Spectral summaries. For a nonzero $\mathbf{w} \in \mathbb{R}_{\geq 0}^m$ define the exponential entropy

$$\text{H}_{\text{exp}}(\mathbf{w}) = \exp\left(-\sum_{i=1}^m \tilde{w}_i \log \tilde{w}_i\right), \quad \tilde{w}_i = w_i / \sum_j w_j, \quad (2)$$

with $0 \log 0 = 0$. The five summaries this paper studies are

$$E_1(\mathbf{H}) = \lambda_1/V, \quad \text{ER}(\mathbf{H}) = \text{H}_{\text{exp}}(\lambda_1, \dots, \lambda_d), \quad \text{RankMe}(\mathbf{H}) = \text{H}_{\text{exp}}(\sigma_1, \dots, \sigma_d), \quad (3)$$

$$\rho(\mathbf{H}) = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{\langle \mathbf{H}_i, \mathbf{H}_j \rangle}{\|\mathbf{H}_i\| \|\mathbf{H}_j\|}, \quad \text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\|\mathbf{X}^\top \mathbf{Y}\|_F^2}{\|\mathbf{X}^\top \mathbf{X}\|_F \|\mathbf{Y}^\top \mathbf{Y}\|_F}. \quad (4)$$

Here $\mathbf{H}_i$ is the i -th row of $\mathbf{H}$, ρ is the anisotropy of Ethayarajh (2019), and CKA is linear centered kernel alignment (Kornblith et al., 2019) for column-centered $\mathbf{X} \in \mathbb{R}^{n \times d_1}$ and $\mathbf{Y} \in \mathbb{R}^{n \times d_2}$. Effective rank (Roy & Vetterli, 2007) and RankMe (Garrido et al., 2023) apply the same functional to different inputs: the training-dynamics literature feeds H_{exp} normalized eigenvalues and the self-supervised-learning literature feeds it normalized singular values. We keep both conventions and label them separately, because Section 3 shows the choice decides how much damage one direction can do.

The projection. Define

$$\mathbf{C} = \mathbf{H}(\mathbf{I}_d - \mathbf{s}\mathbf{s}^\top), \quad \mathbf{s} = \mathbf{v}_1. \quad (5)$$

We call $\mathbf{C}$ the *residual* and $\mathbf{s}$ the *leading direction*. Removing $\mathbf{s}$ from a matrix of embeddings is the All-but-the-Top postprocessing of Mu & Viswanath (2018) with one component; Li et al. (2020) and Rajaei & Pilehvar (2021) use related corrections to improve downstream sentence similarity. This paper uses the projection as a measurement control rather than as a way to improve a downstream score.

Sink strength. For a corpus of B sequences of length T , let α be the mean norm of the hidden states at sequence position 0 divided by the mean norm at all other positions. An attention sink at this layer makes α large (Xiao et al., 2024; Sun et al., 2024). We also report the *position-0 mass* of the leading direction, the fraction of $\sum_i \langle \mathbf{H}_i, \mathbf{s} \rangle^2$ contributed by rows at position 0.

Models, corpora, and protocol. The training-dynamics measurements use Pythia at 160M, 410M and 1.4B with the 20 released checkpoints from step 0 to step 143,000 (Biderman et al., 2023), and the 13 released stage-1 checkpoints of OLMo-2 1B (OLMo Team et al., 2025). Every checkpoint sees the same 606,208 OpenWebText tokens (Gokaslan & Cohen, 2019) in 1,184 sequences of 512. Three disjoint blocks of 392 sequences, or 200,704 tokens each, serve as independent samples; the remaining 8 sequences are held out for the probe loss. Each tokenizer gets its own array with its own published SHA-256 prefix: 28,552 distinct ids for the Pythia array, 33,846 for the OLMo-2 one, and no duplicated sequences in either. Weights are loaded in float32 and all layers are measured. The median cross-block spread at the peak layers runs from 0.1 to 1.9 percent, with a worst case of 4.0 percent; Figure 2 shades it. Section 6 adds final-checkpoint measurements of GPT-2 and GPT-2-XL (Radford et al., 2019), Pythia-6.9B, and Qwen2.5-1.5B (Yang et al., 2024) on WikiText-2 (Merity et al., 2017) and OpenWebText from an earlier round; Table 9 lists the corpus and token count behind every number in the paper. Total compute is under 40 GPU-hours.

3 One direction fixes the value of a rank statistic

The identity below covers effective rank and RankMe, the two summaries of Section 2 that are exponential entropies of a normalized spectrum. Anisotropy gets an exact decomposition and CKA an asymptotic bound; both are weaker, and Section 6 reports how much of each the direction actually accounts for. Proofs are in Appendix A.

3.1 An entropy split and an ordering

Lemma 1 (Top-entry split). *Let $\mathbf{w} \in \mathbb{R}_{\geq 0}^m$ with $\|\mathbf{w}\|_1 > 0$, let $w_1 = \max_i w_i$, let $t = w_1 / \|\mathbf{w}\|_1 < 1$, and let $\mathbf{w}_{-1}$ be $\mathbf{w}$ with its first entry deleted. Then*

$$H_{\text{exp}}(\mathbf{w}) = e^{h_b(t)} \cdot H_{\text{exp}}(\mathbf{w}_{-1})^{1-t}, \quad h_b(t) = -t \log t - (1-t) \log(1-t). \quad (6)$$

Lemma 1 restates the chain rule for Shannon entropy. Everything below follows from it and from one observation: centering acts on the left and the projection in Equation 5 acts on the right, so $\tilde{\mathbf{C}} = \tilde{\mathbf{H}}(\mathbf{I} - \mathbf{s}\mathbf{s}^\top)$, the covariance of $\mathbf{C}$ equals $\Sigma - \lambda_1 \mathbf{s}\mathbf{s}^\top$, and its spectrum is $\{\lambda_2, \dots, \lambda_d, 0\}$.

Lemma 2 (Share ordering). *Let $p = \lambda_1 / \sum_i \lambda_i$ and $r = \sigma_1 / \sum_i \sigma_i$ with $\sigma_i = \sqrt{n\lambda_i}$. Then*

$$\frac{p}{r} = \frac{\sigma_1 \sum_i \sigma_i}{\sum_i \sigma_i^2} \geq 1, \quad (7)$$

with equality if and only if every nonzero singular value equals σ_1 , that is, the spectrum is flat on its support.

3.2 The identities

Proposition 1 (Leading share). *Let $x = \lambda_1 / \text{tr}(\Sigma - \lambda_1 \mathbf{s}\mathbf{s}^\top)$. Then $E_1(\mathbf{H}) = x/(x+1)$, a strictly increasing bijection from $[0, \infty)$ onto $[0, 1)$. The value of $E_1(\mathbf{H})$ carries exactly the ratio of leading to residual variance, and nothing about the shape of the residual spectrum.*

Proposition 2 (Effective rank). *With $p = E_1(\mathbf{H}) < 1$,*

$$\text{ER}(\mathbf{H}) = e^{h_b(p)} \cdot \text{ER}(\mathbf{C})^{1-p}. \quad (8)$$

Proposition 3 (RankMe). *With $r = \sigma_1 / \sum_i \sigma_i < 1$, $\text{RankMe}(\mathbf{H}) = e^{h_b(r)} \cdot \text{RankMe}(\mathbf{C})^{1-r}$.*

Corollary 1 (Distortion, and which metric it reaches). *Write $D_M = M(\mathbf{C})/M(\mathbf{H})$. Then $D_M = M(\mathbf{C})^t e^{-h_b(t)}$ with t the share belonging to M , so*

$$D_M \leq M(\mathbf{C}), \quad D_M \rightarrow M(\mathbf{C}) \text{ as } t \rightarrow 1, \quad (9)$$

and $D_M > 1$ exactly when $t \log M(\mathbf{C}) > h_b(t)$. At every share a matrix can realize, $t \geq 1/(1 + M(\mathbf{C}))$, so $\partial_t \log D_M \geq 0$ and D_M is nondecreasing in t . Compared at a common base, $r \leq p$ therefore gives $D_{\text{RankMe}} \leq D_{\text{ER}}$. At their own bases the comparison is not settled by the shares, because $\text{RankMe}(\mathbf{C}) \geq \text{ER}(\mathbf{C})$ always.

Corollary 1 settles how to read a distortion figure. A study that removes the leading direction from a $d = 4096$ model and reports a $700\times$ change in effective rank has reported a large p and a residual rank near 10^3 . The same p in a $d = 512$ model cannot produce a factor above a few hundred, because D never exceeds $\text{ER}(\mathbf{C})$. Cross-model comparisons of distortion factors compare model width as much as they compare contamination.

Proposition 4 (Anisotropy). *Let $\hat{\mathbf{u}}_i = \mathbf{H}_i / \|\mathbf{H}_i\|$ and $\beta_i = \langle \hat{\mathbf{u}}_i, \mathbf{s} \rangle$. Then*

$$\rho(\mathbf{H}) = \frac{\|\sum_i \hat{\mathbf{u}}_i\|^2 - n}{n(n-1)} = \frac{(\sum_i \beta_i)^2 + \|\sum_i \hat{\mathbf{u}}_i^\perp\|^2 - n}{n(n-1)}, \quad (10)$$

where $\hat{\mathbf{u}}_i^\perp = \hat{\mathbf{u}}_i - \beta_i \mathbf{s}$. The direction contributes $(\sum_i \beta_i)^2 \leq n^2 \max_i \beta_i^2$ to $\|\sum_i \hat{\mathbf{u}}_i\|^2$, one of two nonnegative terms, so the decomposition bounds its contribution without predicting that it dominates. Deleting that term does not give $\rho(\mathbf{C})$: the projection changes the row norms, and $\rho(\mathbf{C})$ renormalizes by $\|\hat{\mathbf{u}}_i^\perp\| = \sqrt{1 - \beta_i^2}$ rather than by 1. Section 6 measures $\rho(\mathbf{C})$ directly.

Proposition 5 (CKA homogenization). *Let $\mathbf{X} = \mathbf{u}\mathbf{p}^\top + \mathbf{X}_c$ and $\mathbf{Y} = \mathbf{u}\mathbf{q}^\top + \mathbf{Y}_c$ be column-centered with $\mathbf{X}_c^\top \mathbf{u} = \mathbf{Y}_c^\top \mathbf{u} = \mathbf{0}$, and set $A = \|\mathbf{u}\|^2 \|\mathbf{p}\|^2$, $B = \|\mathbf{u}\|^2 \|\mathbf{q}\|^2$. Then*

$$1 - \text{CKA}(\mathbf{X}, \mathbf{Y}) = O(A^{-1/2} + B^{-1/2}), \quad (11)$$

with constants depending only on $\|\mathbf{X}_c^\top \mathbf{Y}_c\|_F$, $\|\mathbf{X}_c^\top \mathbf{X}_c\|_F$, and $\|\mathbf{Y}_c^\top \mathbf{Y}_c\|_F$. Two layers sharing one high-norm rank-one term have $\text{CKA} \rightarrow 1$ whatever their residuals do.

Proposition 6 (Specificity of the leading direction). *Removing the k -th eigendirection, $\mathbf{H}_{-k} = \mathbf{H}(\mathbf{I} - \mathbf{v}_k \mathbf{v}_k^\top)$, gives $E_1(\mathbf{H}_{-k}) = \lambda_1 / (V - \lambda_k) \geq E_1(\mathbf{H})$ for $k \geq 2$. For $\mathbf{u}$ drawn uniformly on the unit sphere, $\mathbb{E}[\mathbf{u}^\top \Sigma \mathbf{u}] = V/d$, so a random direction removes $1/d$ of the total variance in expectation.*

3.3 The identity is exact and the ordering is never violated

Figure 1(a) evaluates Equation 8 on every record. The median relative error is 5.3×10^{-16} and the largest is 2.9×10^{-15} , which is the accumulated rounding of a double-precision eigendecomposition. Panel (b) plots

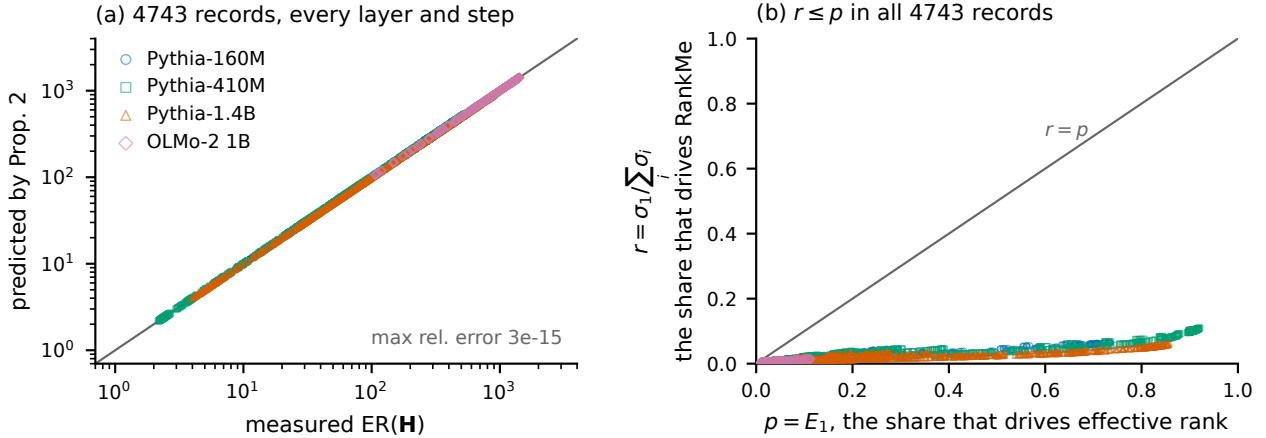


Figure 1: **The value of a rank statistic is fixed by one share and the residual.** (a) Effective rank predicted by Equation 8 against the measured value, on every record we hold: Pythia at 160M, 410M and 1.4B over 20 checkpoints, OLMo-2 1B over 13, all layers of each, three disjoint corpus blocks, and the 300-record sample-size sweep of Table 10. 4,743 points in total. The largest relative error is 2.9×10^{-15} , at the level of double precision. (b) The two shares. Effective rank is driven by $p = E_1$ and RankMe by $r = \sigma_1 / \sum_i \sigma_i$. Lemma 2 forces $r \leq p$; the median ratio is 6.2 and the largest is 20.6. Points near $p = 1$ and $r \approx 0.06$ are the converged Pythia checkpoints, where the same direction destroys effective rank and barely moves RankMe.

Table 1: **The same direction, two statistics, two outcomes.** Peak-sink layer, step 143,000, mean over three disjoint corpus blocks. p and r are the shares of Propositions 2 and 3; r is recovered by inverting the identity. D is the raw-to-residual ratio. The final column is the bound of Corollary 1 for effective rank.

Model	Layer	d	$p = E_1$	r	p/r	D_{ER}	D_{RankMe}	$ER(\mathbf{C})$
Pythia-160M	L4 of 12	768	0.696	0.061	11.5	$32\times$	$1.18\times$	356
Pythia-410M	L9 of 24	1024	0.913	0.104	8.8	$235\times$	$1.45\times$	548
Pythia-1.4B	L5 of 24	2048	0.853	0.058	14.7	$243\times$	$1.24\times$	1023

the two shares against each other. Lemma 2 is never violated, the median p/r is 6.2, and the converged Pythia checkpoints sit at the bottom right, where p approaches 1 while r stays near 0.06.

Table 1 reads the split off the converged checkpoints. At Pythia-1.4B the leading direction holds 85.3 percent of the eigenvalue mass and 5.8 percent of the singular-value mass. Feeding those two shares through Equation 8 and its RankMe analogue gives $243\times$ and $1.24\times$, and both match measurement. What separates the two statistics is the squaring. Squaring moves mass onto the leading entry, so a statistic that squares before it normalizes is the one a single direction can capture.

A reported change in any of these quantities adds nothing to a report of the corresponding share and the residual value. What is left to explain is why the share moves, which is the subject of the next two sections.

4 The leading direction is the attention sink

Section 3 holds for any matrix. It becomes a statement about transformers once we identify $\mathbf{s}$.

Table 2 tracks the direction across checkpoints. At initialization it places a tenth of a percent of its mass on position 0 and α sits near 1.2, so it is ordinary content variance and $E_1(\mathbf{H})$ agrees with $E_1(\mathbf{C})$ to two digits. By the end of training the same direction carries about half its energy on the 1/512 of tokens at position 0, a 250-fold enrichment, and α has reached 19.5 at 160M and 44.7 at 410M. $E_1(\mathbf{C})$ sits between 0.028 and 0.048 at both ends of training in all three models. It rises as high as 0.16 during the early transient

Table 2: **The leading direction migrates onto position 0 during training.** Peak-sink layer, mean over three disjoint blocks of 200,704 OpenWebText tokens. Position 0 is $1/512 = 0.2$ percent of the corpus. The residual share $E_1(\mathbf{C})$ is flat from random initialization to convergence in all three models, so the entire E_1 trajectory lives in one direction.

Model	Layer	α		position-0 mass		$E_1(\mathbf{C})$	
		step 0	step 143k	step 0	step 143k	step 0	step 143k
Pythia-160M	L4	1.23	19.5	0.001	0.441	0.041	0.048
Pythia-410M	L9	1.20	44.7	0.001	0.569	0.036	0.028
Pythia-1.4B	L5	1.22	25.8	0.001	0.498	0.038	0.033
OLMo-2 1B	L3	1.01	1.18 [†]	0.008	0.002 [†]	0.015	0.008 [†]

[†]OLMo-2 stage-1 step 36,000, which is 76B tokens and about 2 percent of its run.

Table 3: **Removing the leading direction is not removing a direction.** Effective rank at the peak-sink layer, step 143,000, same corpus. The random column is the mean and standard deviation over five uniformly drawn unit directions. Proposition 6 predicts the direction of the PC2 and PC3 columns; for a direction that is not an eigenvector it gives the expected variance removed and not the spectrum, so the random column is measurement.

Model	Layer	baseline	remove PC1	remove PC2	remove PC3	remove random
Pythia-160M	L4	11.0	352.1	10.5	10.6	10.9 ± 0.01
Pythia-410M	L9	2.3	539.8	2.3	2.3	2.3 ± 0.00

of Section 5 and comes back down. A residual that stays this diffuse is what makes Equation 5 a clean split; the projection is not rebalancing two components of comparable size.

Attention concentrates on the same position. Norm and variance at position 0 identify a massive-activation direction; the name attention sink also asks for attention mass. Over 19 Pythia-70M checkpoints, mean attention to position 0 at L3 rises from 0.013 to 0.135, a factor of ten, and ranks with α at Spearman 0.725 ($p = 4.5 \times 10^{-4}$). Both are flat until step 1,000 and both climb afterwards.

Where the identification holds. The identification needs a spectral gap. In GPT-2 the ratio σ_1/σ_2 sits near 14.7 across L2 to L4 and falls to 1.1 at L0 and 3.0 at L11. In Qwen2.5-1.5B it reaches 154 at L2. Where the gap is near 1 the leading direction is not separated from content directions and $\mathbf{s}$ should not be read as the sink. We report the gap alongside every layer-level claim and treat $\sigma_1/\sigma_2 > 3$ as the condition under which the identification is meaningful.

Only the first direction matters. Table 3 removes PC1, PC2, PC3 and five random unit directions from the same matrices. At Pythia-410M L9 removing PC1 takes effective rank from 2.3 to 539.8. Removing PC2, PC3 or a random direction leaves it at 2.3. Table 3 comes from a separate pipeline with its own forward pass and eigendecomposition. On the corpus block the two share it reproduces Table 2 to seven digits at Pythia-160M L6, against a spread of 1 to 5 percent between that table’s own three blocks. The two pipelines share the algebra for eigendirection removal, so this checks the data path rather than the identity; the random-direction columns are computed the harder way, by forming the projected covariance and re-diagonalizing it.

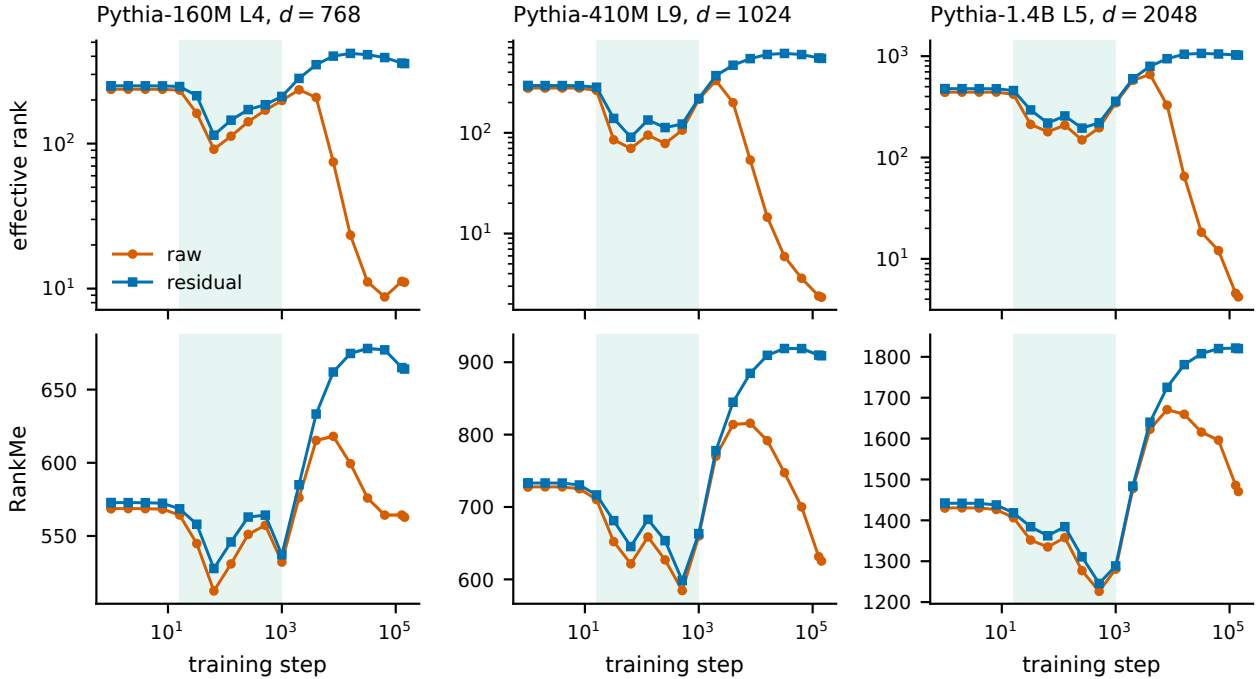


Figure 2: **The collapse is the sink; the early transient is not.** Peak-sink layer of each model, 20 released checkpoints, mean over three disjoint blocks of 200,704 OpenWebText tokens with the block spread shaded. Shading in green marks steps 16 to 1,000. **Top:** effective rank. Raw falls by $21\times$ to $119\times$; the residual dips early, recovers, and ends above where it started. **Bottom:** RankMe on the same matrices. Neither curve collapses, and the raw curve turns down after the sink forms while the residual keeps rising.

Table 4: **Trajectory endpoints at the peak-sink layer.** Step 0 to step 143,000, mean over three disjoint corpus blocks. The last column restricts to the window in which the sink actually grows. Raw effective rank falls by $21\times$ to $119\times$ while the residual rises; raw RankMe changes by a few percent and gets the sign wrong twice.

Model	Layer	effective rank		RankMe		RankMe, 8k to 143k
		raw	residual	raw	residual	
Pythia-160M	L4	237 $\rightarrow$ 11.1	250 $\rightarrow$ 356	-1.0%	+15.9%	-8.9%
Pythia-410M	L9	279 $\rightarrow$ 2.3	295 $\rightarrow$ 548	-14.1%	+24.0%	-23.3%
Pythia-1.4B	L5	442 $\rightarrow$ 4.2	478 $\rightarrow$ 1023	+2.8%	+26.2%	-12.0%

5 Sink growth accounts for the reported trajectories

5.1 Two phenomena, one of which survives the projection

Figure 2 tracks the peak-sink layer of each Pythia model, and two features of the raw effective-rank curve have been read as training phases. One is a drop and recovery before step 1,000. The other is a long decline that starts around step 4,000.

The early drop survives the projection. At Pythia-1.4B residual effective rank falls from 478 to 195 between step 16 and step 256 and then recovers, and residual E_1 rises to 0.096 over the same steps before returning below 0.03 by step 2,000. Sink strength never leaves $[1.05, 1.23]$ in that window, so there is no sink to remove. The other two models do the same thing at the same steps. Something in the representations reorganizes here, and projecting out one direction does not undo it.

Table 5: **A synthetic rank-one term walks both metrics across their range.** Pythia-70M step 512 layer 3, starting from residual states with $E_1 = 0.063$ and $\text{ER}(\mathbf{C}) = 139.9$. The last row applies Equation 8 with $\text{ER}(\mathbf{C})$ held at 139.9. Agreement begins at $\alpha = 3$, where the injected term overtakes the largest content direction; below that the leading direction is still content and 139.9 is the wrong reference.

Injected α	0	1	2	3	5	7	10	15	20	30
E_1	0.063	0.062	0.060	0.081	0.197	0.325	0.496	0.688	0.797	0.898
ER measured	139.9	140.8	136.3	124.1	86.8	52.8	24.2	8.7	4.5	2.3
ER from Eq. 8	129.7	129.9	130.4	124.1	86.8	52.8	24.2	8.7	4.5	2.3

The later decline does not survive. Table 4 gives the endpoints. Raw effective rank ends the run at 4.2, 2.3 and 11.1, factors of 105, 119 and 21 below where it started, and the residual ends above where it started in all three models. The sign of the trajectory is opposite in the two readings, at every scale, on the same tokens.

5.2 RankMe does not collapse and still points the wrong way

The bottom row of Figure 2 runs RankMe on the same matrices. Nothing collapses. Raw RankMe moves by a few percent across all of training where raw effective rank moves by $21\times$ to $119\times$, which is what Corollary 1 requires: the RankMe share sits between 0.06 and 0.10 at these layers, and the effective-rank share between 0.70 and 0.91.

A few percent is still enough to invert the reading. The right-hand columns of Table 4 put the raw and residual RankMe trajectories side by side. Two of the three raw curves have the wrong sign, and the third understates the residual trend by a factor of 9.4. The disagreement is concentrated where it should be. Before step 8,000 the two curves are almost on top of each other; afterwards, as the sink grows, raw RankMe turns down by 9 to 23 percent while the residual stays flat or keeps rising. A RankMe trajectory read at face value reports a late decline that is not in the representations.

5.3 Sink strength predicts the leading share

Proposition 1 reduces E_1 to the ratio x of leading to residual variance. A rank-one term of magnitude α contributes variance proportional to α^2 , which predicts $x = \kappa\alpha^2$ and

$$E_1(\mathbf{H}) = \frac{\kappa\alpha^2}{\kappa\alpha^2 + 1}. \quad (12)$$

We fit the single parameter κ over the 20 checkpoints of each model, with α and E_1 taken from the same tokens. The fits give R^2 between 0.939 and 0.971, and the implied x correlates with α^2 at $r \geq 0.978$ in all three. A scalar computed from token norms, blind to the spectrum, accounts for most of a trajectory that has been read as representational reorganization.

5.4 Injecting a sink into residual states reproduces the trajectory

The fits above show that α tracks E_1 . To test the direction of the relation we start from Pythia-70M at step 512, where $\alpha = 1.24$ and the sink has not formed. We remove the leading direction to obtain $\mathbf{C}$ with $E_1 = 0.063$ and $\text{ER} = 140$, then add back a rank-one term $\alpha \mathbf{e}_0 \mathbf{s}^\top$ of controlled magnitude. Table 5 reports the sweep: as α grows from 3 to 30, E_1 rises from 0.08 to 0.90 and effective rank falls from 124 to 2.3, covering the range the raw trajectories traverse over training. The content matrix is fixed, so the entire excursion comes from the injected term. For $\alpha \leq 2$ the injected term is smaller than the largest content direction, so the leading eigenvector is still a content direction and 139.9 is the wrong reference. The scope rule of Section 4 shows up here as an eight percent prediction error.

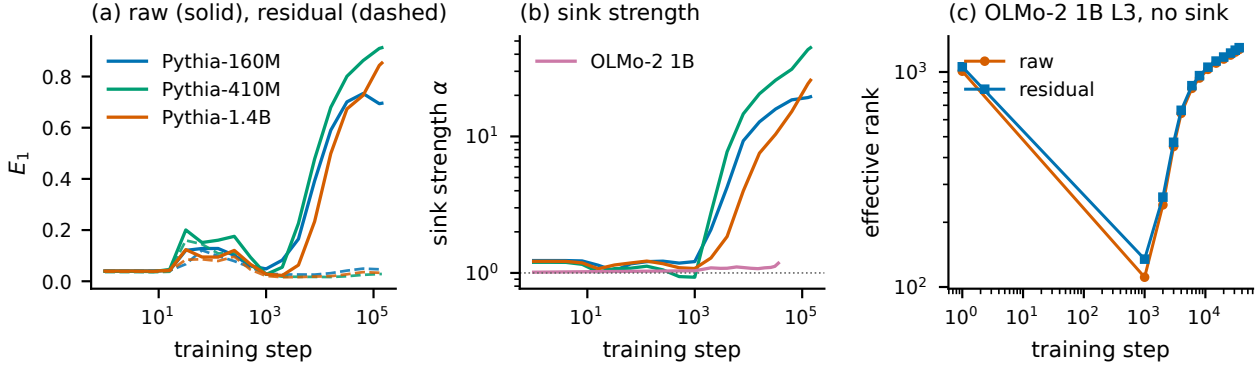


Figure 3: **The raw-to-residual gap is a function of sink strength.** (a) E_1 raw (solid) and residual (dashed) at the peak-sink layer of each Pythia model. The raw curves separate from the residual exactly when α leaves 1. (b) Sink strength. OLMo-2 1B stays below 1.27 across all 17 layers and 13 checkpoints. (c) OLMo-2 1B L3. With no sink, raw and residual effective rank agree to a median of 3.5 percent and both rise. The dip at step 1,000 is the sink-free early transient of Section 5, here at $\alpha = 1.04$.

Table 6: **Five summaries, at the layer where E_1 peaks and at the last layer.** 163,840 OpenWebText tokens, final checkpoints. Raw values on the left of each arrow, residual on the right. “Rogue” is the share of variance in the top five coordinates (Timkey & van Schijndel, 2021); “sink mass” is the fraction of the leading direction’s ℓ_1 mass carried by those five coordinates.

Model	Layer	E_1	Rogue	sink mass	RankMe	Anisotropy
GPT-2 (124M)	L3	93.6% $\rightarrow$ 11.5%	94.4% $\rightarrow$ 18.4%	88.2%	352 $\rightarrow$ 612	0.550 $\rightarrow$ 0.481
	L12	82.2% $\rightarrow$ 48.5%	94.0% $\rightarrow$ 70.0%	56.0%	191 $\rightarrow$ 346	0.829 $\rightarrow$ 0.556
Pythia-70M	L3	62.6% $\rightarrow$ 7.0%	61.9% $\rightarrow$ 7.9%	33.4%	355 $\rightarrow$ 415	0.338 $\rightarrow$ 0.184
	L6	8.2% $\rightarrow$ 4.6%	4.3% $\rightarrow$ 4.3%	2.0%	459 $\rightarrow$ 464	0.982 $\rightarrow$ 0.133

5.5 A model without a sink shows no gap

Every claim above compares a raw quantity with one computed after a projection. A reader may ask how much of the gap the projection itself creates. OLMo-2 1B answers this without an intervention: over its first 76B tokens it never develops a sink. Sink strength stays below 1.273 across all 17 layers and 13 checkpoints, and the position-0 mass of the leading direction stays below 0.012. On those matrices raw and residual effective rank differ by a median of 3.5 percent, and raw effective rank rises monotonically from 111 to 1272 after the early dip. The projection removes a direction in OLMo-2 exactly as it does in Pythia; what changes is that the direction is not carrying anything.

OLMo-2 also reproduces the early transient. Its effective rank falls from 1008 at initialization to 111 at step 1,000 with $\alpha = 1.04$, then climbs for the rest of the measured run. The sink-free rank dip appears in two architecture families.

We do not read this as evidence that OLMo-2 never forms a sink. Step 36,000 is about 2 percent of its schedule, Pythia’s sink at these layers only becomes visible after a comparable fraction, and OLMo-2’s α is drifting up at the last measured checkpoints. The claim supported here is that OLMo-2 has no sink over the interval we measured, and that where there is no sink there is no gap.

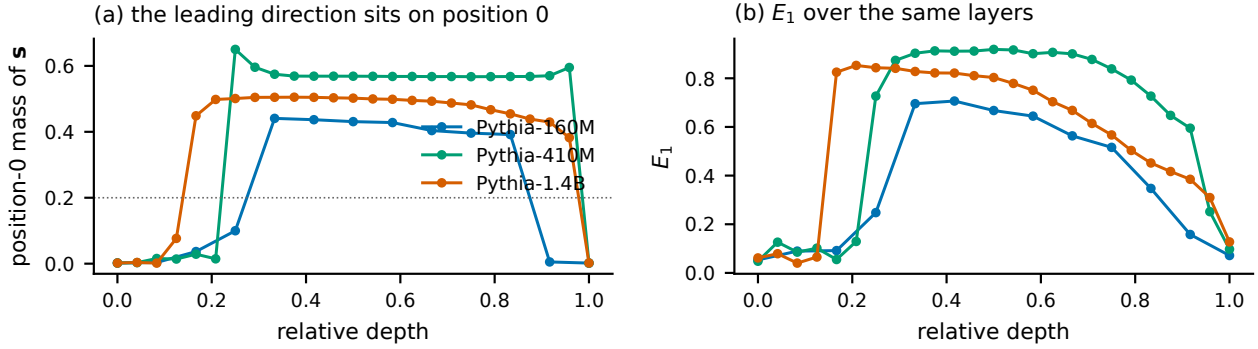


Figure 4: **A contiguous band of layers shares one left factor.** Final checkpoint, mean over three corpus blocks, depth normalized so the three models overlay. (a) Position-0 mass of the leading direction. It holds a narrow range across a contiguous band of layers and collapses at both edges of the band. This is the hypothesis Proposition 5 needs. (b) E_1 over the same layers, for reference. The two profiles have the same support.

6 Scope across layers, scales, and architectures

6.1 Rogue dimensions and the leading direction are the same object

Timkey & van Schijndel (2021) identified a handful of coordinates that dominate cosine similarity in transformer representations and argued that they obscure representational quality. In GPT-2 the top five such coordinates carry 87.3 to 88.6 percent of the leading direction’s ℓ_1 mass at layers L3 through L10, against 3.8 percent at L0 and 20.9 percent at L1. The rogue dimensions of Timkey & van Schijndel (2021) and the attention sink of Xiao et al. (2024) are the same phenomenon in two coordinate systems. Table 6 gives the five summaries at the peak layer of two models.

6.2 Anisotropy responds to the sink but is not driven by it

Proposition 4 splits anisotropy into a sink term and a residual term without asserting which of the two dominates. Measurement says the residual term does. Removing the leading direction at the peak-sink layers moves uncentered anisotropy by 6 to 22 percent, where effective rank on the same matrices moves by $32\times$ to $243\times$. Anisotropy also moves when the sink cannot be responsible. It reaches 0.68 at step 32 in the 160M model and 0.74 at step 16 in the 410M model, with α between 1.0 and 1.2 at both points, then falls to a minimum near step 2,000 and rises again as the sink grows. The late rise is where the sink could contribute, and at the final checkpoint the projection removes 6 to 22 percent of the level. Anisotropy is sink-sensitive; we stop short of calling it sink-driven.

6.3 Raw CKA tracks one shared direction across ten layers

Proposition 5 asks two layers to share one high-norm rank-one term. Figure 4 measures that. At the final checkpoint of each Pythia model the position-0 mass of the leading direction holds a narrow range across a contiguous band of layers: 0.39 to 0.44 over L4 to L10 at 160M, 0.57 to 0.65 over L6 to L23 at 410M, 0.38 to 0.51 over L4 to L23 at 1.4B. At every layer outside a band it is 0.10 or less, and at every layer but one per model it is under 0.05. Eighteen consecutive layers that place close to the same fraction of their leading direction’s energy on the same rows share a left factor, and the band has the shape that raw CKA reports as one indistinguishable block.

Figure 5 shows what that does to CKA in GPT-2. Raw CKA rates L2 through L11 as indistinguishable, and after the projection the same block shows a monotone decay in depth. The off-diagonal mean falls from 0.703 to 0.610 and the largest single change is -0.569 , at the L2-L11 pair that raw CKA rated most similar.

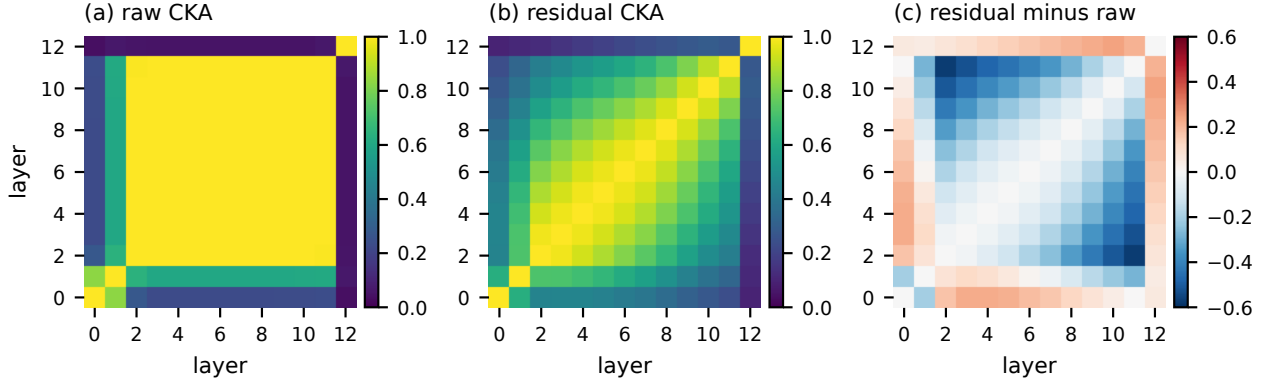


Figure 5: **Raw CKA reports one shared direction.** GPT-2, 22,819 WikiText-2 tokens. (a) Raw CKA marks L2 through L11 as indistinguishable at $\text{CKA} \geq 0.995$. (b) After the projection the same layers show a gradient in depth. (c) The difference. The largest single change is -0.569 between L2 and L11, the pair raw CKA rates most similar.

Table 7: **Contamination at the layer of maximum distortion, final checkpoints.** $D = \text{ER}(\text{C})/\text{ER}(\text{H})$, and the bound is Corollary 1. The two 1.5B models have almost the same bound and differ by $1.8\times$ in D ; Pythia-6.9B has $2.4\times$ the bound of either.

Model	Arch.	Params	Layer	$E_1(\text{H})$	$E_1(\text{C})$	D	bound	$\text{ER}(\text{C})$
GPT-2-XL	GPT-2	1.5B	L18	0.942	0.088	288		518
Qwen2.5-1.5B	GQA	1.5B	L4	0.998	0.037	532		546
Pythia-6.9B	GPT-NeoX	6.9B	L8	0.949	0.024	733		1298

In Pythia-70M the off-diagonal mean rises from 0.360 to 0.444, with the largest change $+0.474$ at L4-L6. The direction of the aggregate change is not stable across models; the layer-pair structure is what changes.

6.4 Larger models produce larger distortion figures, partly by construction

Table 7 reports the layer of maximum distortion for three larger models. Pythia-6.9B reaches $D = 733$, GPT-2-XL reaches 288, and Qwen2.5-1.5B reaches 532. Corollary 1 caps each at the residual rank in the last column, and the cap is what separates the scales: 518 and 546 for the two 1.5B models against 1298 for Pythia-6.9B. A headline that contamination worsens with scale is in part a statement about model width. Within a width the cap does no work, and the two 1.5B models differ by $1.8\times$ in D on caps that differ by $1.05\times$; there the difference is E_1 , which reaches 0.998 in Qwen2.5-1.5B at L4 while GPT-2-XL peaks at 0.942. Qwen2.5-1.5B also carries the largest spectral gap we measured, 154 at L2, with a position-0 norm ratio of 113.

7 What the leading direction does not do

The direction carries no probeable information where it dominates. Table 8 reports linear probing on Pythia-70M. At L3 and L4, where the leading direction holds 63 to 65 percent of the variance, removing it changes accuracy by 0.04 and 0.06 points. Across L0 through L5 the largest change is 0.24 points. A direction that holds most of the variance and none of the next-token information a linear readout can use is a scaffold for the forward pass rather than a store of content, which is the reading of Barbero et al. (2025) and Zhang et al. (2025).

L6, the output of the last block, is the exception and it is instructive. There the leading direction holds 0.4 percent of the variance, the spectral gap implied by Table 6 is 1.4, and removing it raises accuracy by 12.13 points. By the scope rule of Section 4 that direction is not the sink. We report the number because it shows

Table 8: **Removing the leading direction does not remove probeable information.** Pythia-70M, next-token identity prediction over the 500 most frequent types, 33,603 samples from 500 WikiText-2 passages, 3 seeds. “Random” is the mean over 5 random unit directions.

Layer	Sink var	Raw	Residual	Δ residual	Δ random
L0	1.5%	28.21 ± 0.03	28.40 ± 0.08	+0.19	−0.02
L1	2.5%	36.20 ± 0.25	35.96 ± 0.18	−0.24	+0.02
L2	7.0%	39.20 ± 0.30	39.06 ± 0.42	−0.14	−0.02
L3	64.7%	40.24 ± 0.28	40.29 ± 0.28	+0.04	−0.00
L4	62.7%	42.24 ± 0.18	42.31 ± 0.20	+0.06	+0.03
L5	25.3%	43.85 ± 0.06	43.74 ± 0.08	−0.11	−0.03
L6	0.4%	30.92 ± 0.49	43.05 ± 0.28	+ 12.13	+0.45

Table 9: **Evidence behind each claim.** Every number in the paper traces to one row. Data files and the script that regenerates the figures and `numbers.txt` ship with the submission.

Claim	Models	Corpus	Tokens	Ckpts
Identity, ordering (Fig. 1)	Pythia 160M/410M/1.4B, OLMo-2	OWT	$3 \times 200,704$	20, 13
Sink identification (Tab. 2)	same	OWT	$3 \times 200,704$	20, 13
Dynamics (Fig. 2, Fig. 3)	same	OWT	$3 \times 200,704$	20, 13
PC specificity (Tab. 3)	Pythia 160M/410M	OWT	200,704	final
Sample size (Tab. 10)	Pythia-1.4B	OWT	8K to 524K	3
Five metrics (Tab. 6)	GPT-2, Pythia-70M	OWT	163,840	final
CKA (Fig. 5)	GPT-2, Pythia-70M	WikiText-2	22,819	final
Scale (Tab. 7)	GPT-2-XL, Pythia-6.9B	WikiText-2	25,600	final
	Qwen2.5-1.5B	WikiText-2	200,000	final
Injection (Tab. 5)	Pythia-70M L3	WikiText-2	200,000	step 512
Probing (Tab. 8)	Pythia-70M	WikiText-2, 500 psg.	33,603	final
Position-0 perplexity	Pythia-410M	12 sentences	≈ 150	final

OWT: OpenWebText, SHA-256 prefix `66f376629b477962`, 28,552 distinct token ids, no duplicated sequences.

the projection doing something real at a layer where the sink identification fails, and we do not attribute the gain to the sink.

The sink direction has no causal privilege at position 0. We measured perplexity on Pythia-410M after removing the leading direction from the position-0 residual stream at one layer at a time, and compared against removing a random direction of the same magnitude, over 5 trials and 12 evaluation sentences. Removing the leading direction raises perplexity by up to $49\times$ over a baseline of 20.4, averaging $10.2\times$ across the 24 layers. The matched-random control averages $24.1\times$ and does more damage than the leading direction at 21 of the 24 layers. Position 0 is fragile to a perturbation in any direction. The heading of this paragraph is the finding: whatever the sink does for the forward pass, a matched-magnitude random direction does more damage, so this experiment gives the sink no special causal status.

Decisions that use relative layer ordering survive the contamination. On Pythia-6.9B we ranked layers by Block Influence, the cosine distance between a layer’s input and output residual stream, and pruned the 8 lowest. Pruning by raw scores gives perplexity 31.27 and pruning by residual scores gives 33.81, against 16.23 for the full model. The two sets overlap in 7 of 8 layers. Distilling GPT-2 into a 6-layer student gives perplexity 5.57 for even layer spacing, 6.31 for raw CKA matching, and 6.01 for residual CKA matching. Cross-model CKA between GPT-2 and Pythia-70M rises from a maximum of 0.016 to 0.116 after the projection, so the small cross-model similarity is not an artifact of a shared sink. The contamination distorts absolute values of spectral summaries and leaves the ordering of layers largely intact.

Table 10: **Half of the summaries still move with sample size.** Pythia-1.4B L5 at four corpus sizes, each the leading prefix of the corpus, so the three smaller ones are nested inside the largest and the largest spans all three sample blocks. Entries are the largest deviation from the value at $n = 524,288$. Ratio-type quantities converge; entropy-based ranks do not, and rise monotonically with n .

	step 512	step 8,000	step 143,000
E_1 (raw)	8.9%	4.8%	0.5%
α	2.0%	4.2%	0.8%
anisotropy (raw)	16.0%	2.7%	1.6%
ER(H)	13.0%	23.4%	7.7%
ER(C)	12.8%	23.5%	27.1%
RankMe(H)	11.1%	8.9%	11.5%

8 Scope and limitations

Effective rank has not converged in sample size. Table 10 resamples Pythia-1.4B L5 at four corpus sizes. E_1 , α , and anisotropy settle to within 2 percent by $n = 524,288$; effective rank and RankMe do not, and both rise monotonically with n . At the final checkpoint ER(**C**) moves from 756 to 1036 over the four sizes. This is the sample-spectrum bias of a 2048-dimensional covariance estimated from finitely many tokens: small eigenvalues are underestimated at low n , so an entropy-based rank approaches its limit from below. Two consequences. Absolute effective ranks in this paper are values at a stated n and not properties of a layer. Comparisons are valid because every point in a given figure uses the same n , and the raw-to-residual ratio at step 143,000 is $194\times$ at $n = 8,192$ and $246\times$ at $n = 524,288$, so the conclusion does not turn on the choice.

The OLMo-2 control covers 2 percent of a run. Step 36,000 is 76B tokens. Pythia’s sink at these layers becomes visible only after a comparable fraction of its schedule, and OLMo-2’s α is rising at the last measured checkpoint. Section 5 claims that OLMo-2 has no sink over the measured interval, which is what a control needs, and not that OLMo-2 never forms one.

One protocol change across the paper. The dynamics results use 200,704 OpenWebText tokens per block. The GPT-2, Pythia-70M, Pythia-6.9B, and Qwen2.5 measurements come from an earlier round with 22,819 to 200,000 tokens on WikiText-2 or OpenWebText, listed per row in Table 9. Given Table 10, effective ranks are not comparable across those rows, and we compare only within a row.

The position-0 perplexity experiment is small. It uses 12 hand-written sentences on one model, which is enough to show that the matched-random control is not smaller than the treatment and not enough to characterize the causal role of the sink.

The projection is linear and applies at one layer at a time. We remove one direction from one layer and recompute summaries at that layer. We do not track how the removal propagates through subsequent layers, and we do not claim that a nonlinear correction would give the same residual.

The identities assume the leading direction is removed. Equations 8 and 9 hold when the projected direction is the top eigenvector. Table 5 shows what happens when it is not: below $\alpha = 3$ the prediction fails by 8 percent. In practice we check $\sigma_1/\sigma_2 > 3$ before reading **s** as the sink.

We do not evaluate whether phases exist. Our result is that the metrics used to detect them are determined by one direction. A study that reports phases in the residual quantities is unaffected by anything here.

9 Related work

Spectral summaries as descriptions of training. Li et al. (2025) track RankMe and a spectral exponent across pretraining and post-training in OLMo and Pythia and report three phases. Garrido et al. (2023) introduced RankMe as a label-free proxy for downstream quality in self-supervised learning, Roy & Vetterli (2007) defined effective rank, Ethayarajh (2019) measured anisotropy in contextual embeddings, and Raghu et al. (2021) used CKA to compare vision transformers with convolutional networks. Kulkarni et al. (2026) trained 108 models and found that effective rank tracks batch size and weight decay more than it tracks performance, a confound in the training configuration rather than in the estimator. Our result is about the estimator. It also says which of these statistics is the safer choice. RankMe absorbs a 1.2 to 1.5 factor from the sink where effective rank absorbs 32 to 243, so a study that must report one number should report RankMe together with its residual.

Attention sinks. Xiao et al. (2024) named the attention sink and used it for streaming inference; Darcet et al. (2024) added register tokens to vision transformers for the same reason. Sun et al. (2024) characterized massive activations and Gu et al. (2025) showed that sinks emerge early and strengthen with scale. Barbero et al. (2025) argued that sinks limit over-mixing, Cancedda (2024) related sink signals to the unembedding spectrum, and Zhang et al. (2025) showed that one or two sink tags explain a large share of PCA variance in attention outputs. Table 2 adds the trajectory of the position-0 mass of the leading eigenvector, which starts at 0.1 percent and reaches half.

The closest prior result. Queipo-de Llano et al. (2025) prove that massive activations force effective rank down at a layer, and read the resulting compression valley as a functional stage of the forward pass. Proposition 2 sharpens that from an inequality at one layer to an identity, and Lemma 2 adds which statistics inherit it. The consequence for training-dynamics measurement follows: given the share and the residual rank, effective rank has no remaining degree of freedom to carry. Where Queipo-de Llano et al. (2025) ask what sinks do inside a forward pass, we ask what they do to a measurement taken across checkpoints.

Removing a dominant direction. Mu & Viswanath (2018) remove the top principal components of word embeddings and improve similarity benchmarks; Li et al. (2020) and Rajaei & Pilehvar (2021) pursue the same goal with normalizing flows and clustering. Those papers optimize a downstream score. We use the same operation as a control and report the residual next to the raw value rather than in place of it. Timkey & van Schijndel (2021) identified the rogue dimensions that Section 6 matches to the leading direction.

10 Conclusion

Effective rank and RankMe factor into a binary-entropy term and a power of the residual value, so a study that reports either has reported a share and a residual and no third thing. The share is not the same for the two statistics, and it is never the larger one for RankMe; that ordering is what decides how much damage a single direction can do. In Pythia-1.4B the direction takes 85 percent of the eigenvalue mass against 6 percent of the singular-value mass, and distorts the two by $243\times$ and $1.24\times$ respectively. It is the attention sink, and it gets there during training: by the end of the run about half its energy sits on the 1/512 of tokens at position 0, while a model that has not yet grown one shows raw and residual metrics a median of 3.5 percent apart. Projecting it out leaves a residual that grows over training, and a single transient before step 1,000 that the sink cannot account for.

Some of this translates into reporting practice. Effective rank should be quoted next to E_1 and $ER(\mathbf{C})$, which determine it, and next to the spectral gap, which tells a reader whether the direction that was removed is the sink at all. Distortion factors need the residual value that bounds them and the token count they were measured at, given that entropy-based ranks have not converged at half a million tokens. If only one rank statistic can be reported, RankMe is the better choice, and it should still be reported with its residual: a $1.24\times$ offset was enough to invert two of our three trajectories. The implementation is one line, $H_{\text{res}} = H - (H @ s)[: , \text{None}] * s$, with s the top right singular vector of the centered H .

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A Proofs

A.1 Lemma 1

Proof. Write $\tilde{w}_i = w_i/\|\mathbf{w}\|_1$, so $\tilde{w}_1 = t$ and $\sum_{i \geq 2} \tilde{w}_i = 1-t$. Let $q_i = \tilde{w}_i/(1-t)$ for $i \geq 2$ be the normalization of $\mathbf{w}_{-1}$. Then

$$-\sum_i \tilde{w}_i \log \tilde{w}_i = -t \log t - \sum_{i \geq 2} (1-t) q_i [\log(1-t) + \log q_i] = h_b(t) + (1-t) \left(-\sum_{i \geq 2} q_i \log q_i \right),$$

using $\sum_{i \geq 2} q_i = 1$. Exponentiating gives Equation 6. The case $t = 1$ makes both sides equal 1 under the convention $0 \log 0 = 0$. $\square$

A.2 Lemma 2

Proof. Write $S_1 = \sum_i \sigma_i$ and $S_2 = \sum_i \sigma_i^2$. Then $p = \sigma_1^2/S_2$ and $r = \sigma_1/S_1$, so $p/r = \sigma_1 S_1/S_2$. Since $\sigma_i \leq \sigma_1$ for every i ,

$$S_2 = \sum_i \sigma_i^2 \leq \sigma_1 \sum_i \sigma_i = \sigma_1 S_1,$$

which gives $p/r \geq 1$. Equality forces $\sum_i \sigma_i(\sigma_1 - \sigma_i) = 0$, and every summand is nonnegative, so $\sigma_i \in \{0, \sigma_1\}$ for each i . A flat spectrum with k nonzero entries gives $p = r = 1/k$, so equality holds for every such spectrum and not only for $k = 1$. $\square$

A.3 Spectrum of the residual

Lemma 3. *With $\mathbf{C}$ as in Equation 5 and $\mathbf{s} = \mathbf{v}_1$, the covariance of $\mathbf{C}$ is $\mathbf{\Sigma} - \lambda_1 \mathbf{s} \mathbf{s}^\top$ and its eigenvalues are $\lambda_2 \geq \dots \geq \lambda_d \geq 0$ together with one extra zero.*

Proof. Centering is left multiplication by $\mathbf{I}_n - \frac{1}{n} \mathbf{1} \mathbf{1}^\top$ and the projection is right multiplication by $\mathbf{I}_d - \mathbf{s} \mathbf{s}^\top$, so the two commute and $\bar{\mathbf{C}} = \bar{\mathbf{H}}(\mathbf{I}_d - \mathbf{s} \mathbf{s}^\top)$. Hence

$$\frac{1}{n} \bar{\mathbf{C}}^\top \bar{\mathbf{C}} = (\mathbf{I} - \mathbf{s} \mathbf{s}^\top) \mathbf{\Sigma} (\mathbf{I} - \mathbf{s} \mathbf{s}^\top) = \mathbf{\Sigma} - \lambda_1 \mathbf{s} \mathbf{s}^\top,$$

where the second equality uses $\mathbf{\Sigma} \mathbf{s} = \lambda_1 \mathbf{s}$ and $\mathbf{s}^\top \mathbf{s} = 1$. In the eigenbasis of $\mathbf{\Sigma}$ this matrix is diagonal with entries $0, \lambda_2, \dots, \lambda_d$. $\square$

A.4 Propositions 1, 2, 3 and Corollary 1

Proof. By Lemma 3, $\text{tr}(\mathbf{\Sigma} - \lambda_1 \mathbf{s} \mathbf{s}^\top) = V - \lambda_1$, so $x = \lambda_1/(V - \lambda_1)$ and $x/(x+1) = \lambda_1/V = E_1(\mathbf{H})$, which proves Proposition 1; the map $x \mapsto x/(x+1)$ is strictly increasing with inverse $p \mapsto p/(1-p)$.

For Proposition 2 apply Lemma 1 to $\mathbf{w} = (\lambda_1, \dots, \lambda_d)$. Its largest entry is λ_1 and $t = \lambda_1/V = E_1(\mathbf{H}) = p$. The deleted vector $\mathbf{w}_{-1} = (\lambda_2, \dots, \lambda_d)$ is the nonzero part of the spectrum of the residual covariance, and appending the extra zero of Lemma 3 leaves H_{exp} unchanged, so $H_{\text{exp}}(\mathbf{w}_{-1}) = \text{ER}(\mathbf{C})$. Substituting into Equation 6 gives Equation 8. Proposition 3 is the same argument with $\mathbf{w} = (\sigma_1, \dots, \sigma_d)$ and $t = \sigma_1/\sum_i \sigma_i$, since $\sigma_i = \sqrt{n \lambda_i}$ preserves the ordering and the residual singular values are $\sigma_2, \dots, \sigma_d$.

For Corollary 1, divide: $D_M = M(\mathbf{C})/M(\mathbf{H}) = M(\mathbf{C})^{1-(1-t)} e^{-h_b(t)} = M(\mathbf{C})^t e^{-h_b(t)}$. Both factors are at most $M(\mathbf{C})$ and 1 respectively for $t \in (0, 1)$ and $M(\mathbf{C}) \geq 1$, so $D_M \leq M(\mathbf{C})$, with equality in the limit $t \rightarrow 1$ where $h_b(t) \rightarrow 0$. Taking logarithms, $D_M > 1$ if and only if $t \log M(\mathbf{C}) > h_b(t)$.

For the monotonicity, $\partial_t \log D_M = \log M(\mathbf{C}) - \log \frac{1-t}{t}$. Applying $H_{\text{exp}}(\mathbf{w}) \geq \|\mathbf{w}\|_1/\|\mathbf{w}\|_\infty$ to $\mathbf{w}_{-1}$ gives $M(\mathbf{C}) \geq (1-t)/t$ at any share a matrix realizes, so the derivative is nonnegative there and D_M is non-decreasing in t . Comparing two statistics at one base M then gives $D \leq D'$ whenever the first share is the smaller, which with Lemma 2 orders RankMe below effective rank. Their own bases are not equal: $\text{RankMe}(\mathbf{C}) \geq \text{ER}(\mathbf{C})$, since squaring a spectrum concentrates it. The spectrum $(100, 1, 10^{-4} \times 999)$ has $p = 0.989$, $r = 0.476$, $\text{ER}(\mathbf{C}) = 2.54$ and $\text{RankMe}(\mathbf{C}) = 723$, giving $D_{\text{ER}} = 2.37$ against $D_{\text{RankMe}} = 11.5$. The ordering of shares alone therefore does not order the distortions. $\square$

A.5 Proposition 4

Proof. Let $\hat{\mathbf{u}}_i = \mathbf{H}_i / \|\mathbf{H}_i\|$ and $S = \sum_i \hat{\mathbf{u}}_i$. Then $\|S\|^2 = \sum_{i,j} \langle \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j \rangle = n + \sum_{i \neq j} \langle \hat{\mathbf{u}}_i, \hat{\mathbf{u}}_j \rangle$, and $\rho(\mathbf{H})$ is the average of the off-diagonal terms, which gives the first equality. Decomposing $\hat{\mathbf{u}}_i = \beta_i \mathbf{s} + \hat{\mathbf{u}}_i^\perp$ with $\langle \mathbf{s}, \hat{\mathbf{u}}_i^\perp \rangle = 0$ gives $\|S\|^2 = (\sum_i \beta_i)^2 + \|\sum_i \hat{\mathbf{u}}_i^\perp\|^2$, the second equality. If $\beta_i = \beta$ for all i then the first term is $n^2 \beta^2$ and $\rho \geq (n^2 \beta^2 - n) / (n(n-1)) \rightarrow \beta^2$ for large n . $\square$

A.6 Proposition 5

Proof. Since $\mathbf{X}_c^\top \mathbf{u} = \mathbf{0}$ and $\mathbf{Y}_c^\top \mathbf{u} = \mathbf{0}$, the cross terms vanish and $\mathbf{X}^\top \mathbf{Y} = \|\mathbf{u}\|^2 \mathbf{p}\mathbf{q}^\top + \mathbf{X}_c^\top \mathbf{Y}_c$. Write $c_{xy} = \|\mathbf{X}_c^\top \mathbf{Y}_c\|_F$, $c_{xx} = \|\mathbf{X}_c^\top \mathbf{X}_c\|_F$, $c_{yy} = \|\mathbf{Y}_c^\top \mathbf{Y}_c\|_F$, and note $\|\mathbf{p}\mathbf{q}^\top\|_F = \|\mathbf{p}\| \|\mathbf{q}\|$, so $\|\|\mathbf{u}\|^2 \mathbf{p}\mathbf{q}^\top\|_F = \sqrt{AB}$. The triangle inequality gives

$$\|\mathbf{X}^\top \mathbf{Y}\|_F^2 \geq (\sqrt{AB} - c_{xy})^2 \quad (\sqrt{AB} \geq c_{xy}), \quad \|\mathbf{X}^\top \mathbf{X}\|_F \leq A + c_{xx}, \quad \|\mathbf{Y}^\top \mathbf{Y}\|_F \leq B + c_{yy}.$$

Hence

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) \geq \frac{AB(1 - c_{xy}/\sqrt{AB})^2}{AB(1 + c_{xx}/A)(1 + c_{yy}/B)} = 1 - O(A^{-1/2} + B^{-1/2}),$$

expanding to first order in $1/\sqrt{A}$ and $1/\sqrt{B}$. Since $\text{CKA} \leq 1$ always, the two-sided statement follows. $\square$

A.7 Proposition 6

Proof. Replacing $\mathbf{s}$ by $\mathbf{v}_k$ in Lemma 3 gives a residual covariance with spectrum $\{\lambda_i\}_{i \neq k}$ plus a zero, whose largest entry is λ_1 for $k \geq 2$ and whose trace is $V - \lambda_k$. Hence $E_1(\mathbf{H}_{-k}) = \lambda_1 / (V - \lambda_k) \geq \lambda_1 / V$. For $\mathbf{u}$ uniform on the sphere, $\mathbb{E}[\mathbf{u}\mathbf{u}^\top] = \mathbf{I}_d/d$, so $\mathbb{E}[\mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}] = \text{tr}(\boldsymbol{\Sigma})/d = V/d$. $\square$

B Additional measurements

Cross-block spread. Median relative standard deviation over the three disjoint 200,704-token blocks, at the peak-sink layers, across all 20 checkpoints: $\text{ER}(\mathbf{H})$ 0.98 to 1.18 percent, $\text{ER}(\mathbf{C})$ 0.73 to 0.87 percent, E_1 1.06 to 1.86 percent, $\text{RankMe}(\mathbf{C})$ 0.08 to 0.16 percent, α 0.20 to 0.25 percent.

Spectral gaps by layer. GPT-2: 1.1, 3.7, 14.7, 14.9, 14.6, 13.4, 12.4, 9.6, 8.4, 5.5, 3.2, 3.0 at L0 through L11. Qwen2.5-1.5B reaches 154.2 at L2 with a position-0 norm ratio of 112.8.

Per-layer contamination at scale. Pythia-6.9B, raw and residual E_1 by layer: L0 0.165/0.035, L4 0.931/0.030, L8 0.949/0.024, L12 0.929/0.023, L16 0.901/0.025, L20 0.853/0.036, L24 0.777/0.047, L28 0.700/0.051, L31 0.075/0.059. In Pythia the first and last layers are uncontaminated. GPT-2 is the exception: its last layer carries $E_1 = 82.2$ percent (Table 6).

Pythia-70M under the earlier protocol. WikiText-2, 22,851 tokens, one block. L3 effective rank falls from 178 to 9.2 raw and moves from 191 to 156 in the residual, with a maximum of 236 at step 4,000. L1, which never develops a sink, moves from 187 to 162 raw and 202 to 190 in the residual. Table 10 shows that effective rank at $n = 22,851$ is far from its large- n value, so we do not compare these numbers with the $n = 200,704$ trajectories of Section 5.

Full injection sweep. Table 5 omits $\alpha = 50$, where $E_1 = 0.961$ and $\text{ER} = 1.43$ against 1.43 predicted by Equation 8.

C Reproduction

The submission ships the experiment scripts, the raw output of every run quoted in the paper, and `build.py`, which regenerates all four figures and writes `numbers.txt` containing every quantity in the manuscript next

to the file it came from. Running `python build.py` takes under a minute and requires no GPU. The measurements use public checkpoints and total under 40 GPU-hours.

An earlier round of our own experiments measured Pythia-160M and Pythia-1.4B training dynamics through a corpus loader whose exception handler silently substituted 200 copies of one template sentence. The resulting effective ranks were near 10 at initialization for a 768-dimensional model, which is the signature of roughly ten distinct token types. Those runs are excluded, every number derived from them is withdrawn, and Sections 4 and 5 report the replacement measurements on the corpus described in Section 2. The corpus is published with its SHA-256 prefix, its distinct-token count, and a check for duplicated sequences; the shipped scripts raise on a corpus-loading failure and refuse to run when a measured effective rank at initialization falls below $d/25$.

Broader Impact Statement

This work identifies a measurement bias in a class of published analyses of transformer training. Its effect is on the accuracy of scientific claims about representation geometry. We see no direct societal risk from the method or from its release.